

4.4 Setup & Installation Guide — CampusCare

Prerequisites

Ensure the following are installed on your machine before proceeding.

Project Structure

CampusCare/

■■■■ backend/ Express.js API server (TypeScript + Prisma)

■ ■■■■ prisma/ Database schema, migrations, seed

■ ■■■■ src/ Source code (controllers, routes, middleware)

■■■■ frontend/ React Native mobile app (Expo SDK 54)

■ ■■■■ src/ Screens, navigation, services, components

■■■■ docs/ Project documentation

Backend Setup

1. Install backend dependencies:

```
cd backend
```

```
npm install
```

2. Configure environment variables in backend/.env

3. Generate Prisma Client:

```
npx prisma generate
```

4. Apply migrations:

```
npx prisma migrate deploy
```

5. Seed database:

```
npm run seed
```

6. Run backend:

```
npm run dev
```

Frontend Setup

1. Install frontend dependencies:

```
cd frontend
```

```
npm install
```

2. Configure BASE_URL in frontend/src/services/api.js

3. Start Expo:

```
npx expo start -c
```

4. Scan QR code using Expo Go app.

Third-Party Services

CampusCare uses Supabase as its hosted PostgreSQL database provider.

Uploaded files are stored locally under backend/uploads/.

Common Issues & Fixes

- Missing dependency → Run: `npx expo install expo-secure-store`
- Login failed → Run `npm run seed`
- Network request failed → Verify BASE_URL and backend server
- Database timeout → Resume Supabase project

Tool Requirements

Tool	Minimum Version	How to Check
Node.js	18.x or higher	<code>node --version</code>
npm	9.x or higher	<code>npm --version</code>
Git	Recent version	<code>git --version</code>
Expo CLI	Installed via npm	<code>npx expo --version</code>
Expo Go	Latest	Install from App Store / Google Play

Useful Backend Commands

Command	Description
npm run dev	Start development server with hot reload
npm run build	Compile TypeScript build
npm start	Run production build
npm run seed	Seed database with test data
npx prisma migrate deploy	Apply migrations